



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

Presentation

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Topic 1: PCM based tunable grating(DBR)

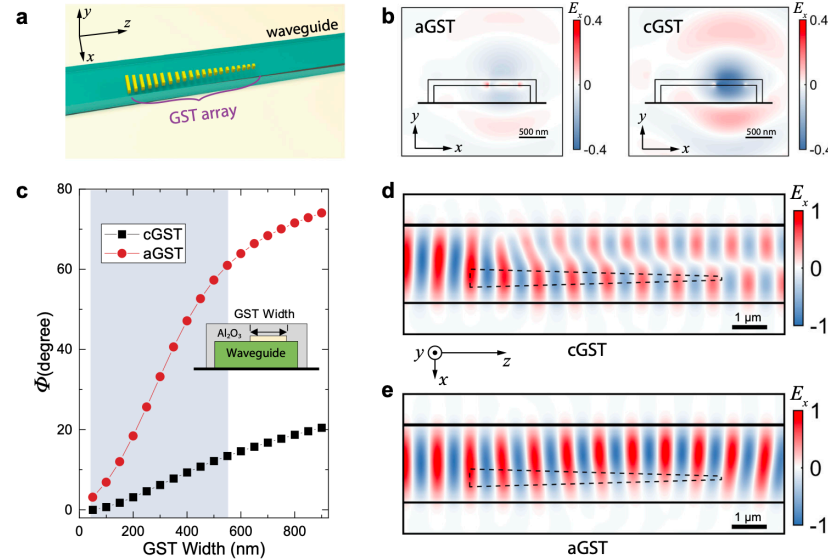
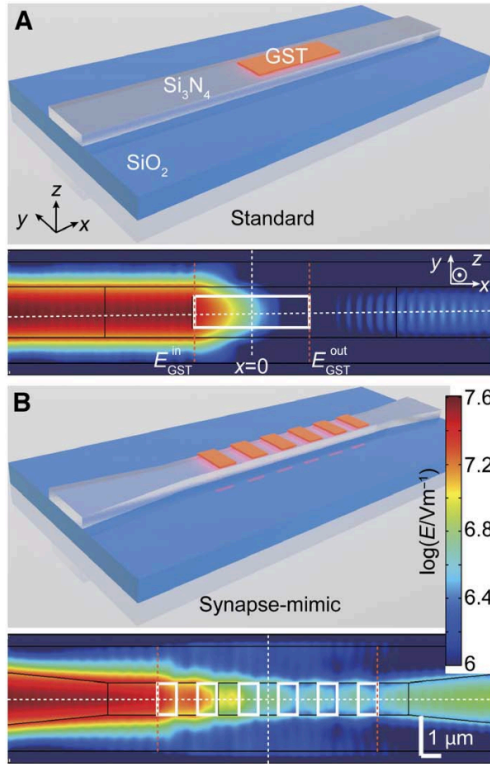


Fig. 1 Design of the phase-gradient metasurface mode converter. **a** 3D illustration of the devices. **b** FDTD simulation of the scattered electric field by one nano-antenna when the GST is in aGST (left panel) and cGST (right panel) phases, respectively, showing the distinctive difference. **c** The phase of the scattered mode as a function of the GST nano-antenna width for cGST and aGST phases. The shaded region indicates the range of antenna widths that are used in the phase gradient metasurface. Inset: cross-sectional view of the structure. **d, e** FDTD simulation results showing effective mode conversion from the TE_0 mode to the TE_1 mode when the GST is in crystalline phase (**d**), but only a small perturbation when the GST is in amorphous phase (**e**).

Cheng et al., Sci. Adv. 3, e1700160 (2017)

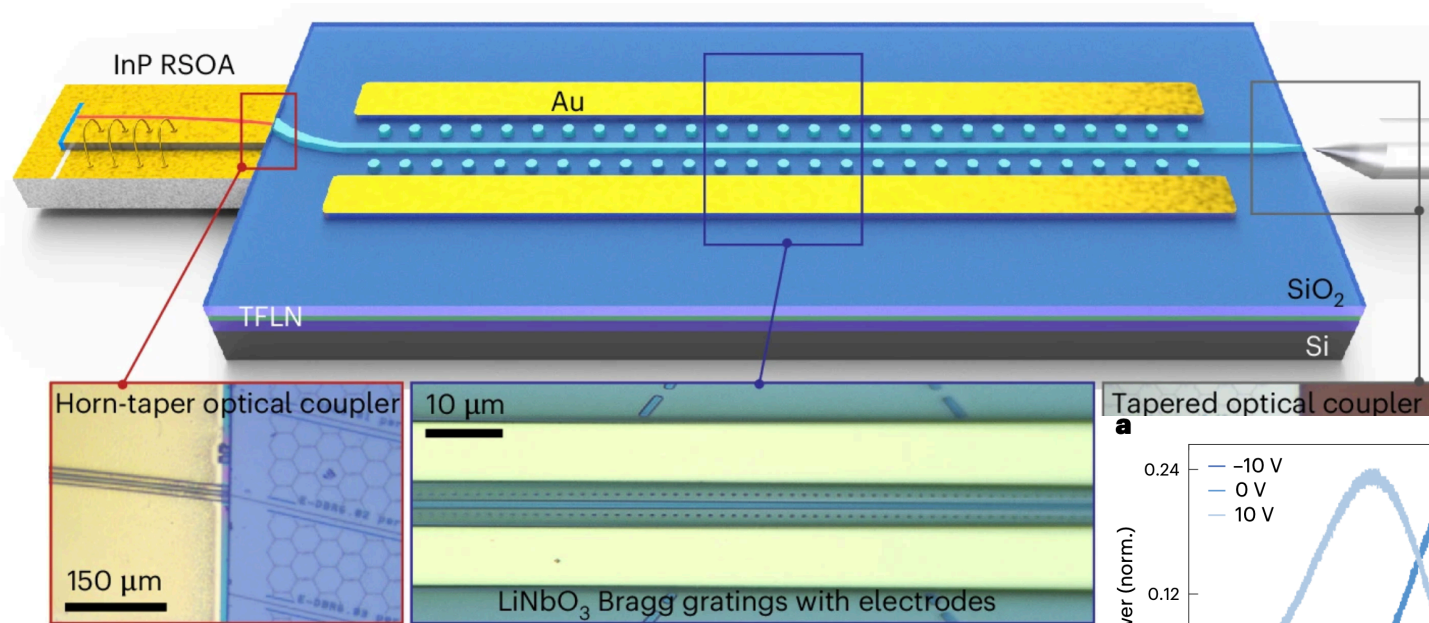
Wu et al., Nat. Commun. 12, 96 (2021)

Loss?c propagat

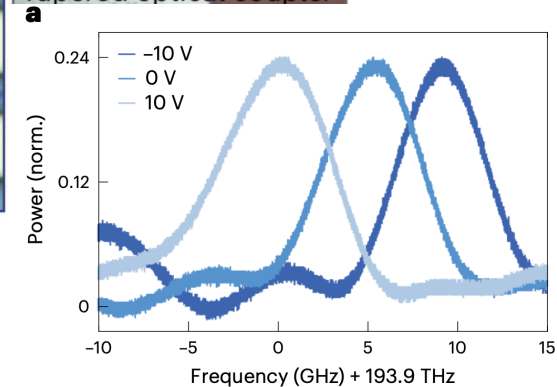


Topic 1: PCM based tunable grating(DBR)

a



Tunable DBR laser based on LN



Nat. Photon. 19 (2025)



Topic 1: PCM based tunable grating(DBR)

$$\lambda_0 = 2(n_H d_H + n_L d_L) \quad \Delta\lambda_{stop} = \frac{4\lambda_0}{\pi} \arcsin\left(\frac{n_H - n_L}{n_H + n_L}\right) \quad R = \left[\frac{1 - (n_H/n_L)^{2N}}{1 + (n_H/n_L)^{2N}} \right]^2$$

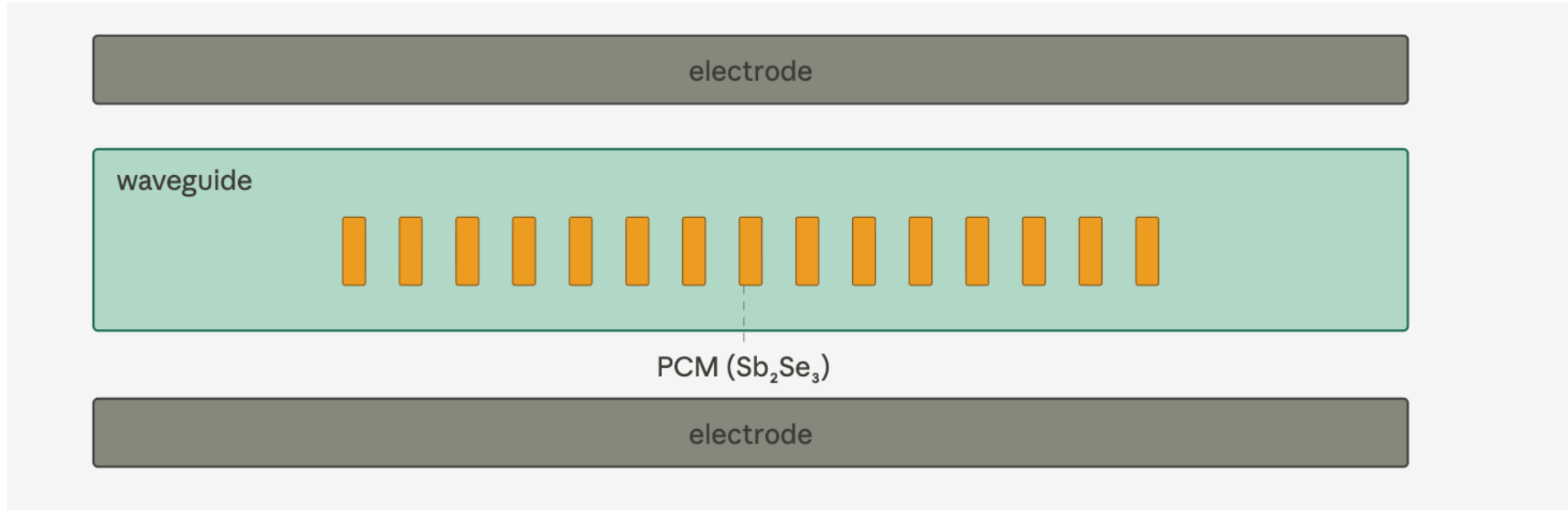
advantages(large n contrast of PCM)

- Wide stopband — high reflectivity maintained over a broad wavelength range
- Fewer layer pairs needed — high reflectivity achieved with small N, enabling compact structures
- Small physical thickness — reduced total device size, easier integration
- Shallow penetration depth — light reflects within the first few pairs, enabling short effective cavity length
- High tolerance to wavelength drift — robust against temperature, angle, and fabrication variations
- Reduced cumulative absorption and scattering — fewer layers mean less total propagation loss
- Broadband operation — supports wide-spectrum sources, femtosecond pulses, and multi-channel systems



Topic 1: PCM based tunable grating(DBR)

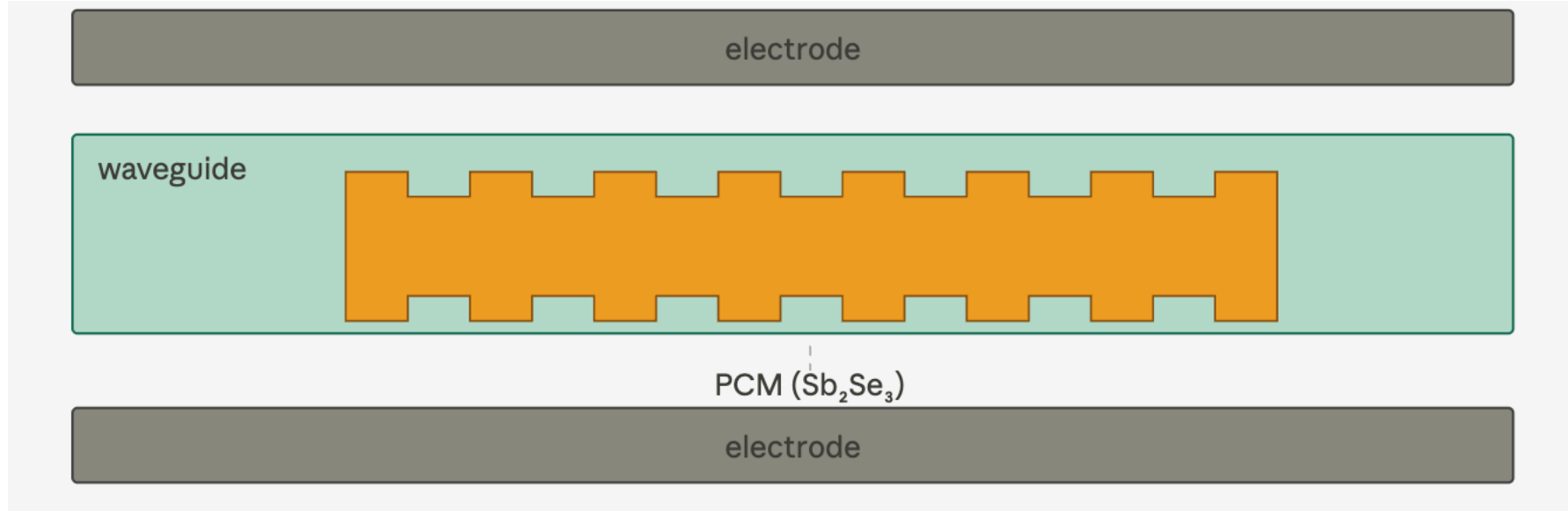
Period, thickness, width



PCM-DBR grating(laser) with tunable n

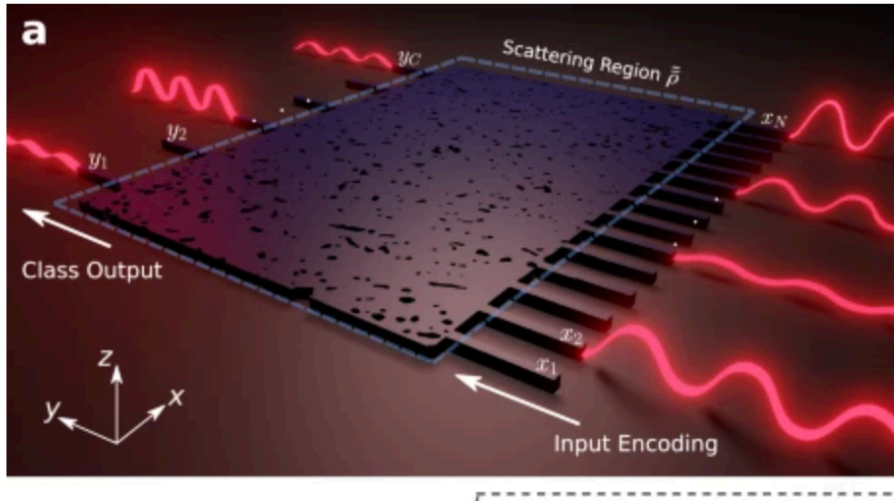


Topic 1: PCM based tunable grating(DBR)

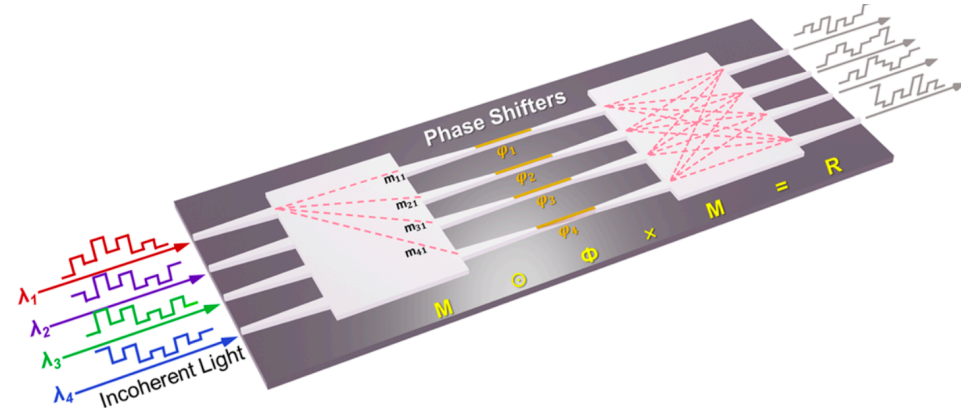


PCM-DBR grating(laser) with tunable n (roughly constant stop band)

Topic 2: PCM based inverse design

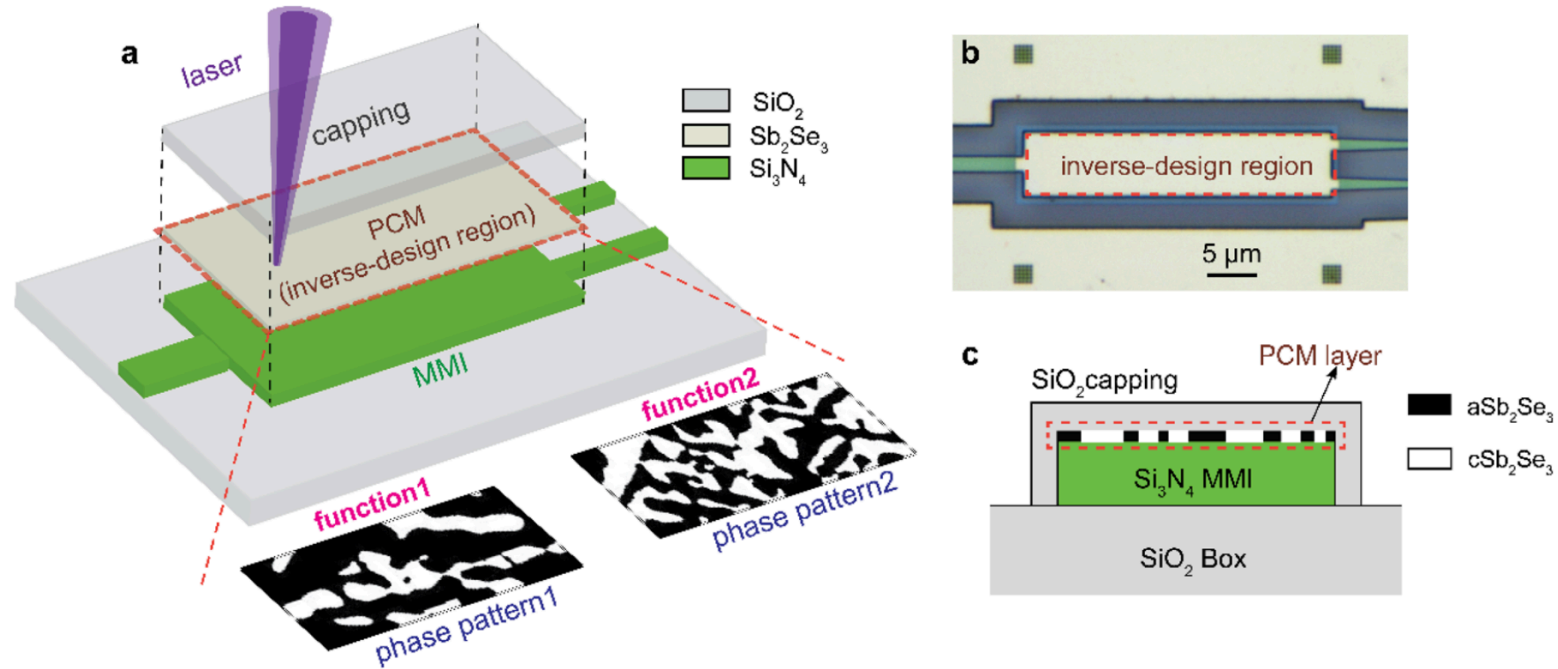


Sved et al., Nat. Commun. (2026)



Meng et al., Nat. Commun. 14, 3000 (2023)

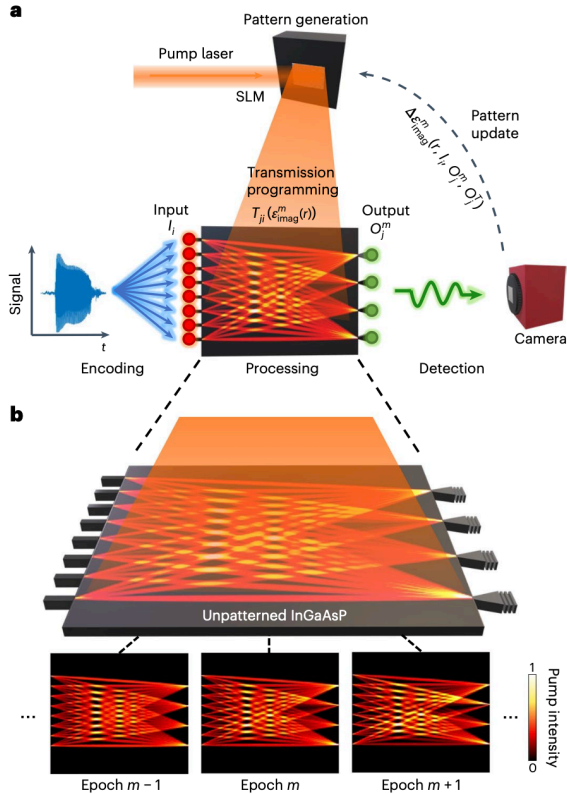
Topic 2: PCM based inverse design



Wu et al., APL Photonics (2025), only used as filter and splitter

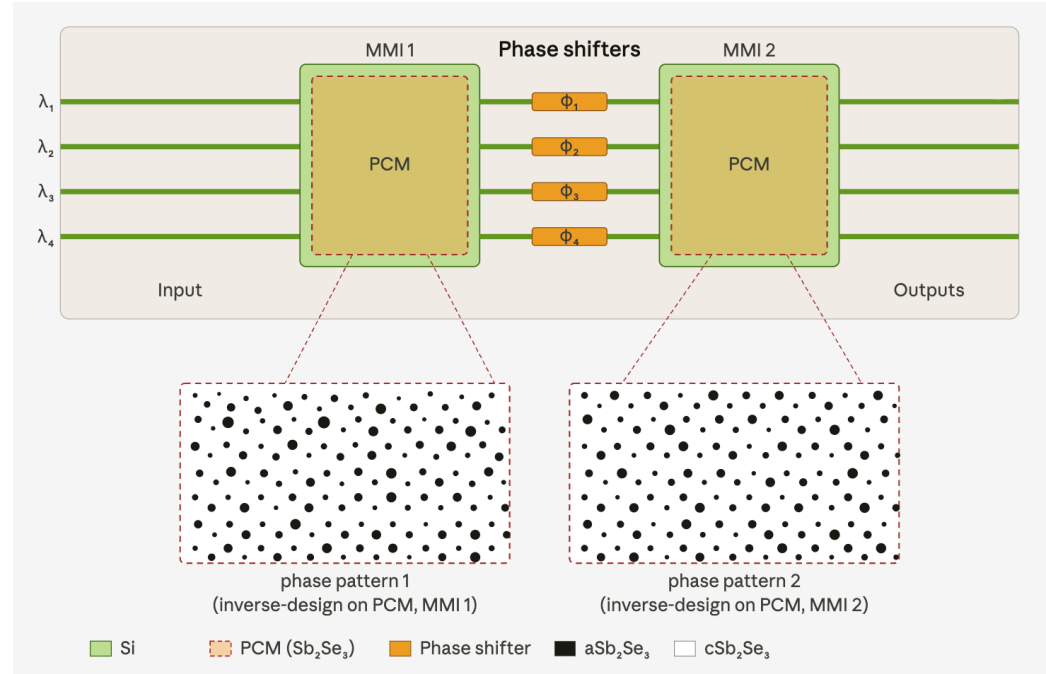


Topic 2: PCM based inverse design



Wu et al., Nat. Photon. (2023)

more compact compared to universal design



Our architecture

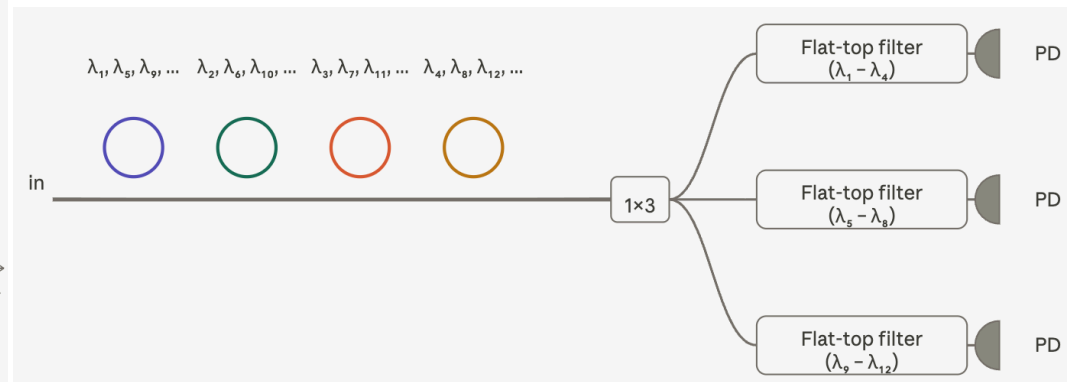
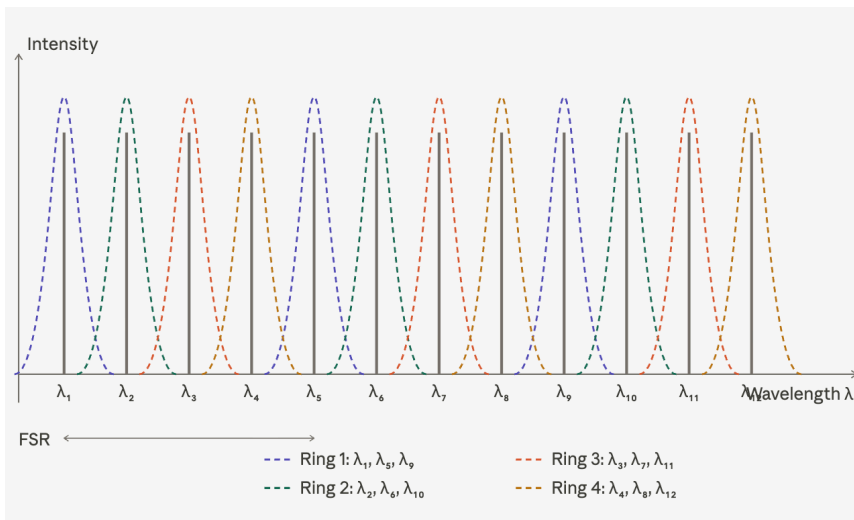
Topic 2: PCM based inverse design

Advantages:

- More compact(compared with MZI mesh)
- More flexible(compared with etching)
- Could also be used both for fine tune



Topic 3: WDM based computing beyond FSR



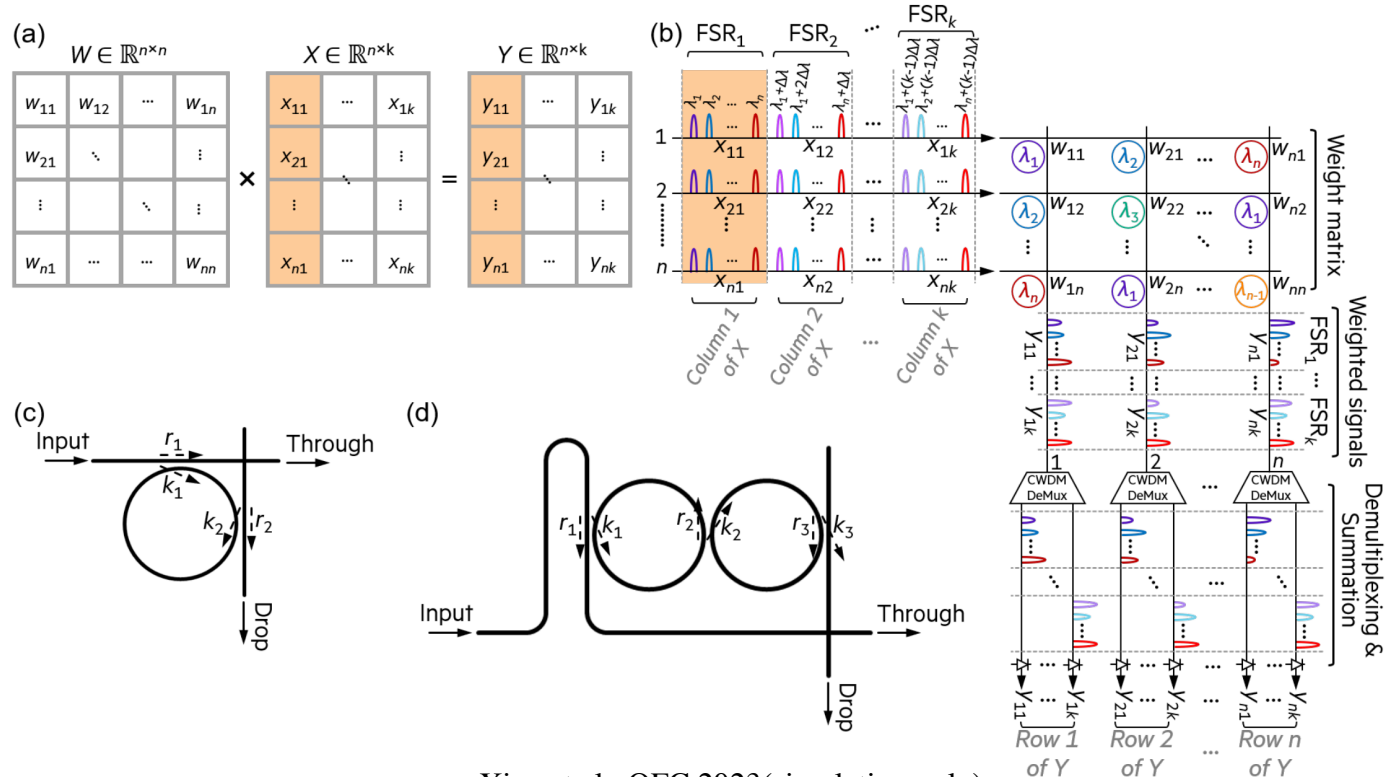
Matrix-matrix multiplication

Key design:

Broadband coupler

Flat-top filter(achromatic phase shifter)

Topic 3: WDM based computing beyond FSR



Xiao et al., OFC 2023(simulation only)

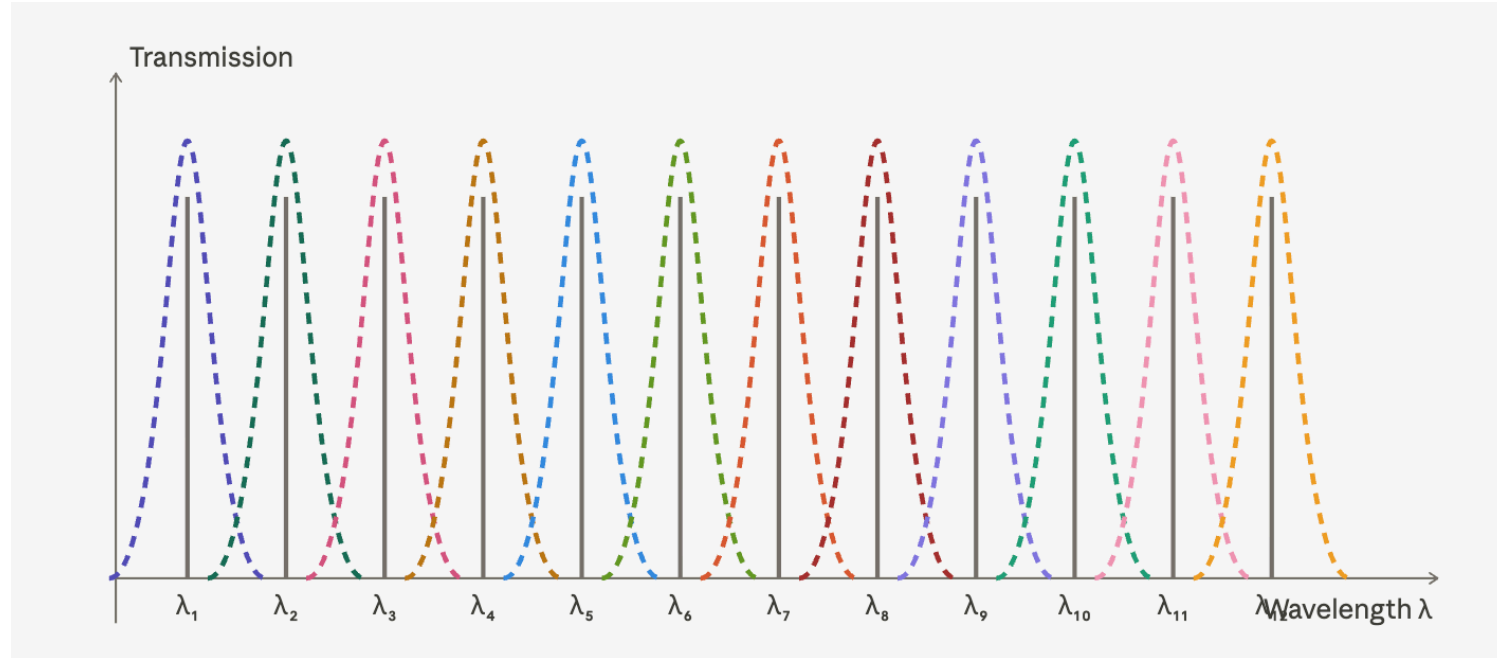


Topic 3: WDM based computing beyond FSR

Advantages:

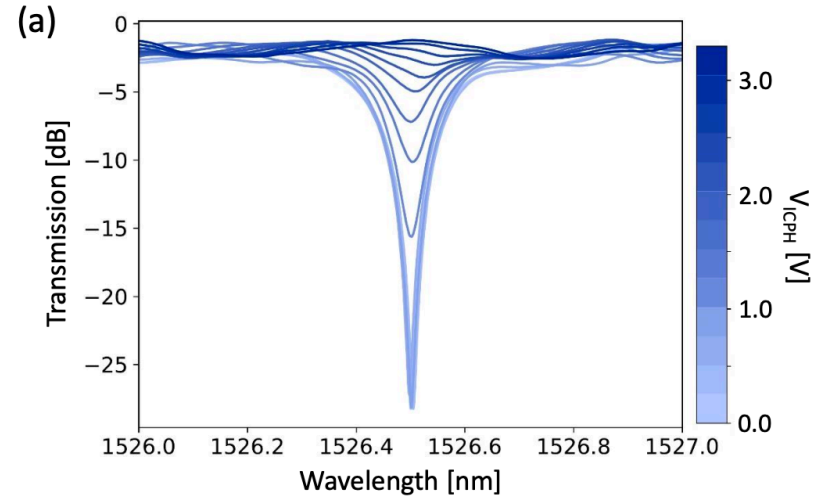
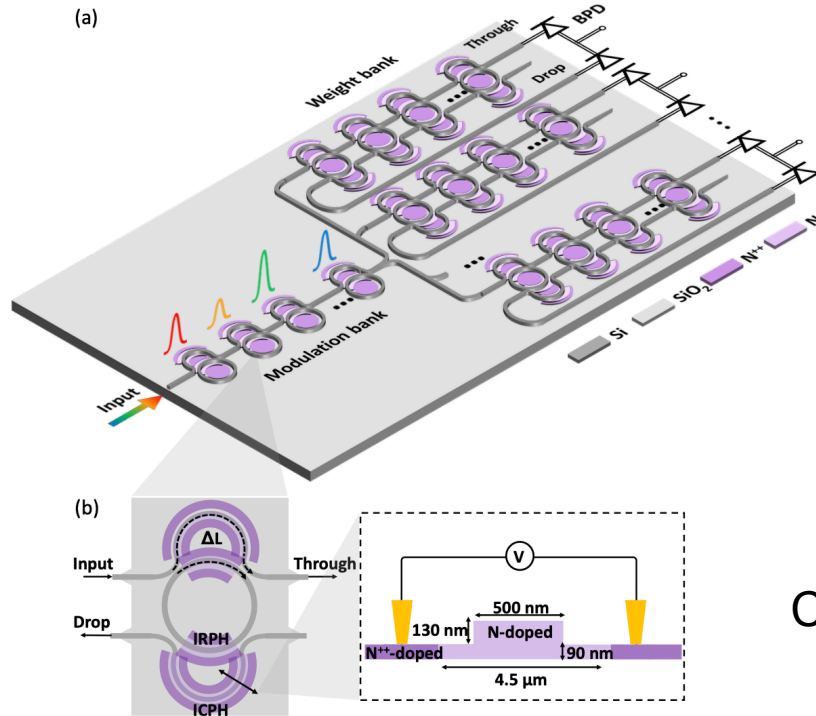
- Large throughput
- Multiple access(different user share same unit)

Topic 4: DWDM architecture based on coupling modulation



Extreme WDM situation(no crosstalk, but shifting peak will bring crosstalk)

Topic 4: DWDM architecture based on coupling modulation

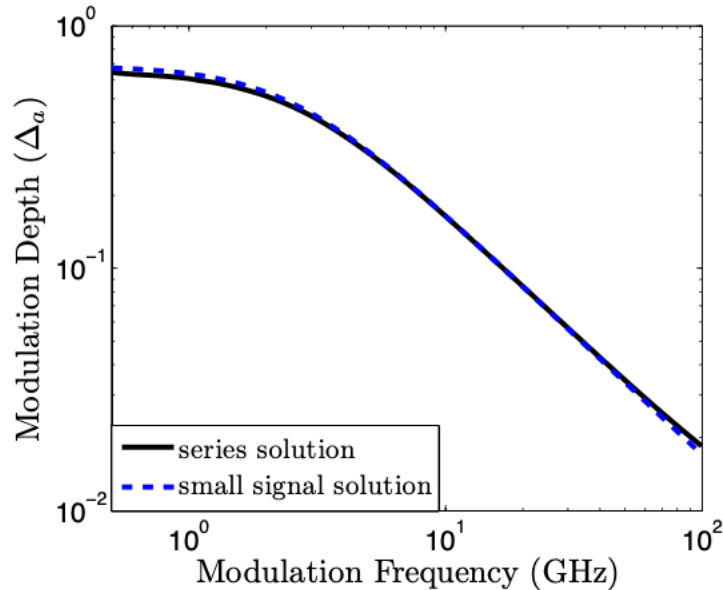


Changing from shifting peak to changing keppa

Luan et al., Preprint/ResearchSquare (2022) (simulation only)

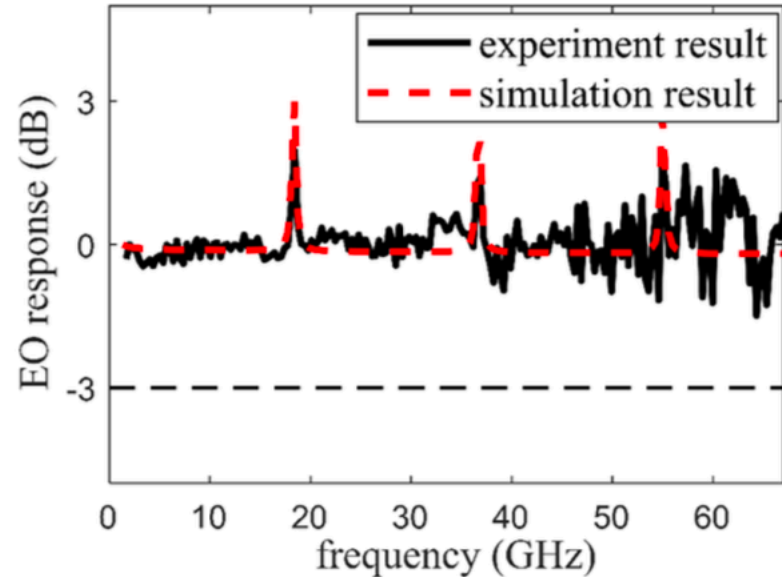
Topic 4: DWDM architecture based on coupling modulation

Free from cavity lifetime



Resonant peak modulation

Sacher & Poon, Opt. Express 16(20) (2008)



Coupling coefficient modulation

Xue et al., Optica 9(10) (2022)

Topic 4: DWDM architecture based on coupling modulation

Advantages:

- Dense channels
- Overcoming the limitation of cavity dynamics



Potential topic

- ◆ Co-designed PIC and metasurface
- ◆ Integrated OAM emitter with tunable topological charge and emitting direction(free space communication and quantum application)
- ◆ Specific PIC for VR display(CGH generation) and integration with metasurface
- ◆ Flat top ring cavity(intrinsic property)
- ◆ Stable Flat top frequency comb
- ◆ Electro-optic neuromorphic and normal computing hardware